SLIIT UNI
THE KNOWLEDGE UNIVERSITY

ME 1060: DYNAMICS

Lecture : Impulse and Momentum

Overview

- Impulse and Momentum
- Linear momentum,
- Impulse of a force
- Conservation of linear momentum

Principle of Linear Impulse and Momentum

In this section we will integrate the equation of motion with respect to time and thereby obtain the principle of impulse and momentum.

Using EoM for a particle of mass m;

$$\sum \boldsymbol{F} = m\boldsymbol{a} = m\frac{d\boldsymbol{v}}{dt}$$

where a and v are both measured from an inertial frame of reference.

Rearranging the terms;

$$\sum \int_{t_1}^{t_2} \boldsymbol{F}\, dt = m \int_{v_1}^{v_2} d\boldsymbol{v} = m\boldsymbol{v_2} - m\boldsymbol{v_1}$$

Linear Momentum

Each of the two vectors of the form L = mv is referred to as the particle's linear momentum.

Since m is a positive scalar, the linear-momentum vector has the same direction as v, and its magnitude mv has units of mass-velocity, e.g., kg m/s.

Linear Impulse

The integral $I = \int F\, dt$ in the equation is referred to as the *linear impulse*.

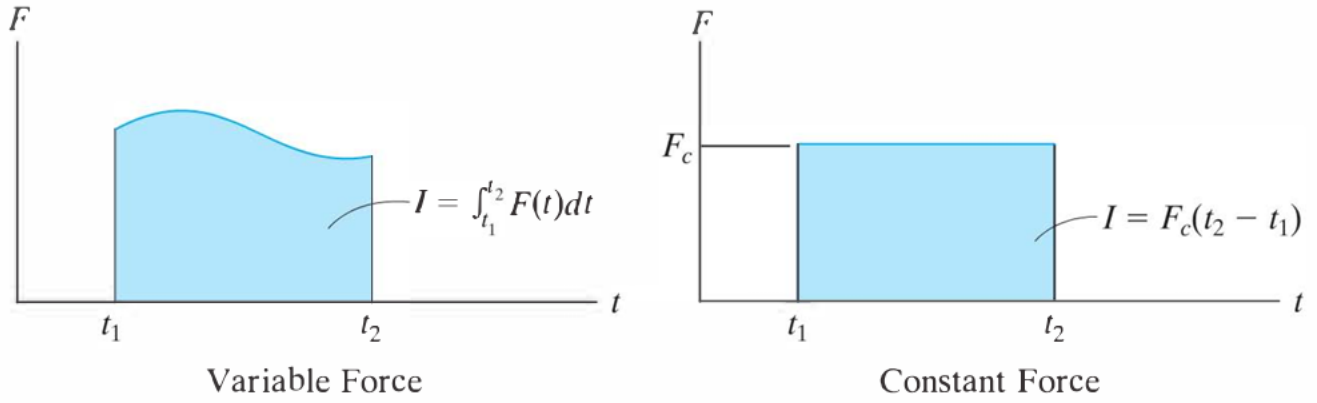
This term is a vector quantity which measures the effect of a force during the time the force acts. Since time is a positive scalar, the impulse acts in the same direction as the force, and its magnitude has units of force-time, e.g., N . s or lb · s.

Linear Impulse

If the force is expressed as a function of time, the impulse can be determined by direct evaluation of the integral.

$$\boldsymbol{I = \int_{t_1}^{t_2} F_c dt = F_c(t_2 - t_1)}$$

Graphically the magnitude of the impulse can be represented by the shaded area under the curve of force versus time



Variable Force

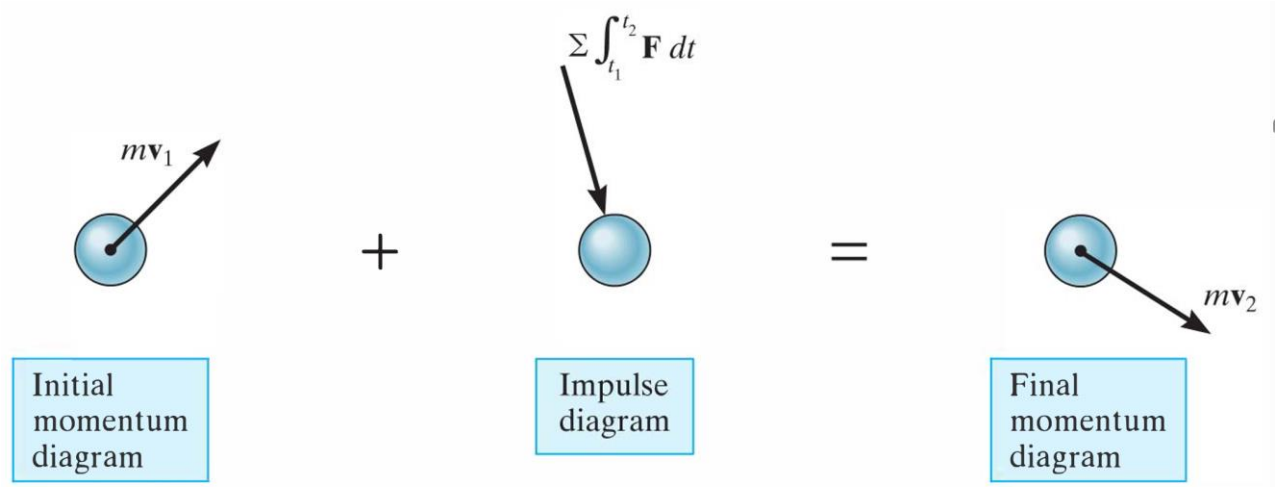
Constant Force

Principle of Linear Impulse and Momentum

We can rearrange the equation:

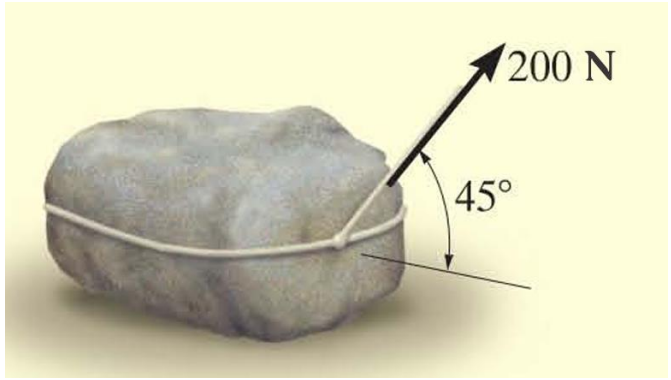
$$m\boldsymbol{v_1} + \sum \int_{t_1}^{t_2} \boldsymbol{F}\, dt = m\boldsymbol{v_2}$$

These three terms are illustrated graphically on the impulse and momentum diagrams



Example 1

The 100-kg stone shown in figure is originally at rest on the smooth horizontal surface. If a towing force of 200 N, acting at an angle of 45°, is applied to the stone for 10 s, determine the final velocity and the normal force which the surface exerts on the stone during this time interval.



Example 1

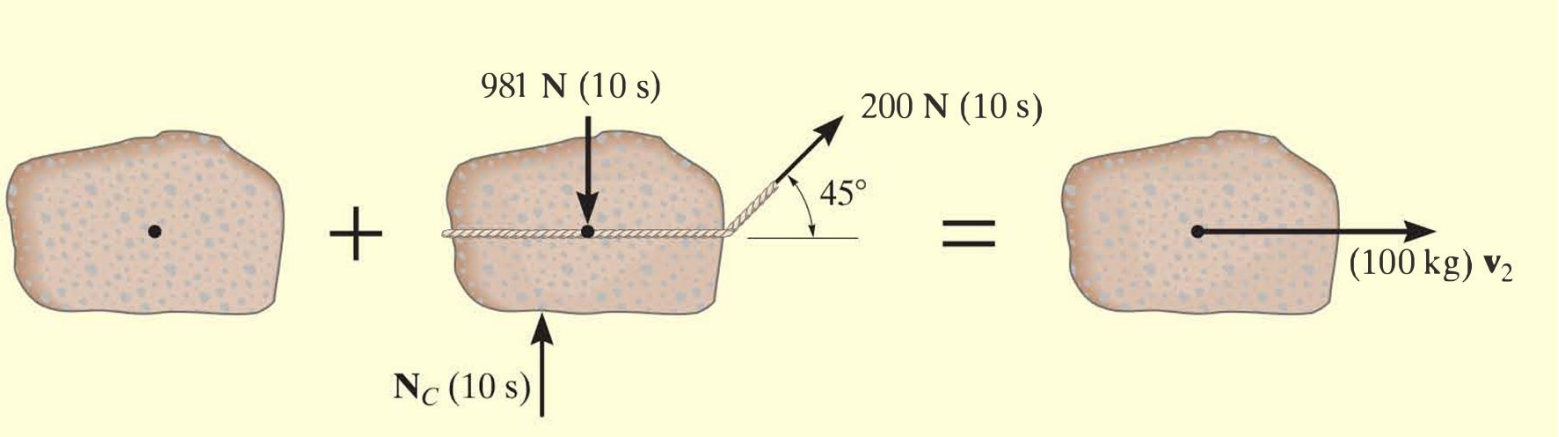
Principle of Impulse and Momentum.

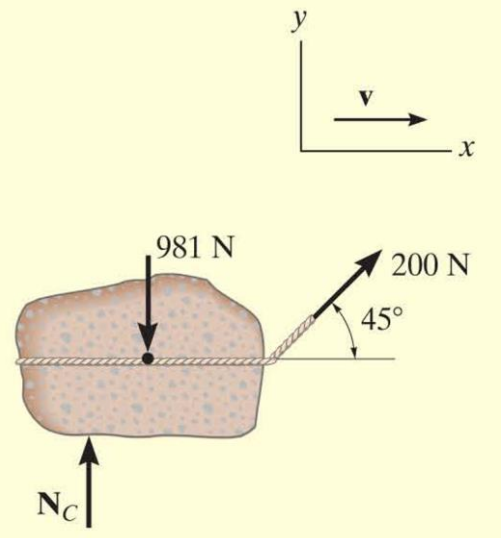
$$m(\boldsymbol{v_x})_1+\sum\int_{t_1}^{t_2}\boldsymbol{F_x}\,dt = m(\boldsymbol{v_x})_2$$

$$0 + 200\cos 45\,(10) = 100\,v_2$$
$$v_2 = 14.1\,m/s$$

$$m(\boldsymbol{v_y})_1+\sum\int_{t_1}^{t_2}\boldsymbol{F_y}\,dt = m(\boldsymbol{v_y})_2$$

$$0 + N_c(10) - 981\,(10) + 200\sin 45\,(10) = 0$$
$$N_c = 840\,N$$





Conservation of Linear Momentum

When the external impulse acting on a system of particles is zero:

$$\sum m_i(v_i)_1 = \sum m_i(v_i)_2$$

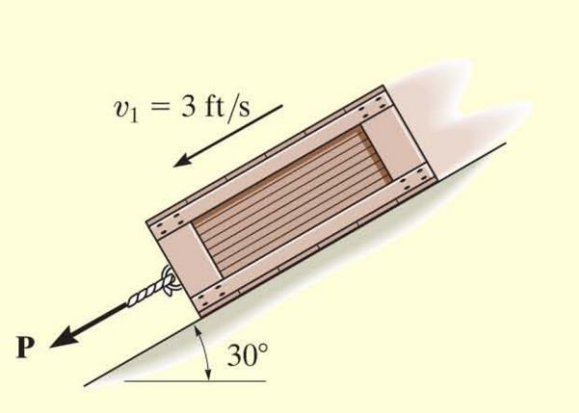


This equation is referred to as the **'Conservation of linear momentum'**

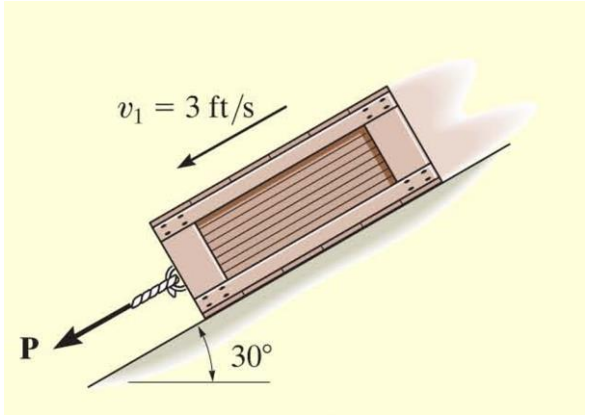
The conservation of linear momentum often applies when particles collide or interact. **Internal impulses** for the system will always cancel out, since they occur in **equal but opposite collinear pairs.**

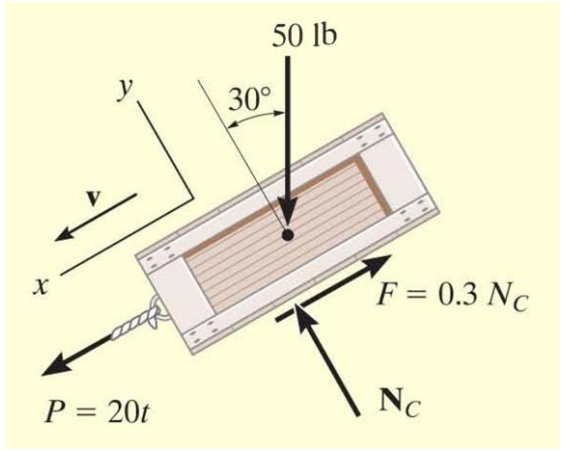
Question 2

The 50-lb crate shown in figure is acted upon by a force having a variable magnitude P = (20t) lb, where t is in seconds. Determine the crate's velocity 2 s after P has been applied. The initial velocity is $v_1 = 3ft/s$ down the plane, and the coefficient of kinetic friction between the crate and the plane is $\mu_k = 0.3$.

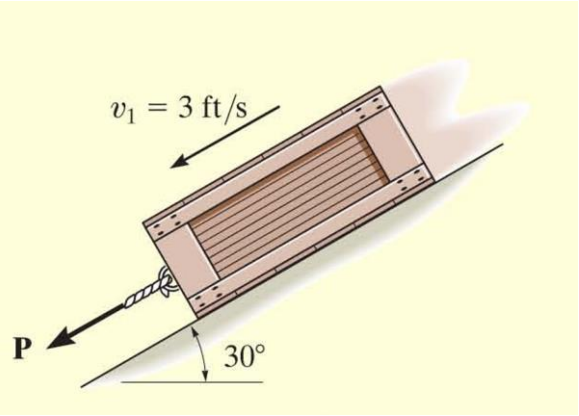


Question 2





Question 2



Principle of Impulse and Momentum. Applying Eqs. 15–4 in the x direction, we have

$$(+\swarrow) \qquad m(v_x)_1 + \Sigma \int_{t_1}^{t_2} F_x\, dt = m(v_x)_2$$

$$\frac{50\text{ lb}}{32.2\text{ ft/s}^2}(3\text{ ft/s}) + \int_0^{2\text{ s}} 20t\, dt - 0.3N_C(2\text{ s}) + (50\text{ lb})\sin 30°(2\text{ s}) = \frac{50\text{ lb}}{32.2\text{ ft/s}^2}v_2$$

$$4.658 + 40 - 0.6N_C + 50 = 1.553v_2$$

The equation of equilibrium can be applied in the y direction. Why?

$$+\nwarrow \Sigma F_y = 0; \qquad N_C - 50\cos 30°\text{ lb} = 0$$

Solving,

$$N_C = 43.30\text{ lb}$$

$$v_2 = 44.2\text{ ft/s}\swarrow \qquad \textit{Ans.}$$

NOTE: We can also solve this problem using the equation of motion. From Fig. 15–5*b*,

$$+\swarrow \Sigma F_x = ma_x;\ 20t - 0.3(43.30) + 50\sin 30° = \frac{50}{32.2}a$$

$$a = 12.88t + 7.734$$

Using kinematics

$$+\swarrow dv = a\, dt; \qquad \int_{3\text{ ft/s}}^{v} dv = \int_0^{2\,s} (12.88t + 7.734)dt$$

$$v = 44.2\text{ ft/s} \qquad \textit{Ans.}$$

Question 3

A group of ten friends, each of mass 50kg, want to test the strength of a trampoline. At their highest jump, each friend is 2m above the trampoline. They all land on the trampoline at the same time and decelerate at a constant rate to a point of zero velocity over a period of 0.5s. The trampoline has a threshold of 5000N, above which it breaks. Do the friends succeed in breaking their trampoline?

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According to the problem statement, we can treat the friends as a single mass of 500kg. We can use the expression for conservation of energy to calculate the velocity as they hit the trampoline.

$$E = U_i + K_i = U_f + K_f$$

If we take the initial state to be when the friends are at their highest point and the final state to be when the friends hit the trampoline, we can rewrite:

$$U_i = K_f$$

Substituting in expressions:

$$mgh_i = \frac{1}{2}mv_f^2$$

Rearranging for velocity:

$$v_f = \sqrt{2gh_i}$$

Plugging in our values:

$$v_f = \sqrt{2(10\frac{m}{s^2})(2m)} = 6.32\frac{m}{s}$$

The statement says that the friends decelerate from this velocity to zero over a period of 0.5 seconds. Therefore, we can write:

$$a = \frac{\Delta v}{t} = \frac{6.32\frac{m}{s}}{0.5s} = 12.65\frac{m}{s^2}$$

We can use this to calculate the force that the trampoline is exerting on the friends:

$$F = ma = (500kg)\left(12.65\frac{m}{s^2}\right) = 6325N$$

This is above the threshold of 5000 N, so the trampoline does break.